MD. SADMAN ANJUM JOARDER

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RESEARCH PROFILE

Mechanical engineer working on carbon capture, utilization, and storage (CCUS), renewable energy systems, and biomass conversion. Author of 23 peer-reviewed publications (16 journal articles, 7 conference papers) with Springer, Elsevier, Wiley, IEEE, and Taylor & Francis, cited more than 230 times (h-index 9); six of the articles appeared in Q1 journals. Trained in experimental hydrothermal conversion of waste biomass, process simulation in Aspen Plus, PCTRAN, and RETScreen, and technoeconomic analysis. Now an Assistant Engineer with Bangladesh's Roads and Highways Department, after a year of teaching in the national technical education system. Applying to PhD programs to continue work on carbon capture and CO_2 conversion.

RESEARCH INTERESTS

Carbon Capture, Utilization, and Storage (CCUS) · CO_2 Separation and Conversion · Renewable Energy Systems and Policy · Solar Photovoltaic Technoeconomics and Recycling · Biomass and Microalgae Conversion (Hydrothermal Carbonization and Liquefaction) · Waste Management and Circular Economy

EDUCATION

Bachelor of Science in Mechanical Engineering *2016–2021*
Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh. CGPA 3.23/4.00.
Thesis: *Hydrothermal Carbonization of Water Hyacinth, Green Algae and Waste Tea Leaves* (Supervisor: Prof. Dr. Mohammad Shahed Hasan Khan Tushar).

Higher Secondary Certificate (HSC), Jhenidah Cadet College, Jashore Board *2014*
GPA 5.00 (Golden); ranked 53rd among all candidates in the board.

Secondary School Certificate (SSC), Jhenidah Cadet College, Jashore Board *2012*
GPA 5.00 (Golden); awarded the Board Secondary Scholarship.

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher, Department of Mechanical Engineering, RUET *2019–2021*

- Converted invasive and waste biomass (water hyacinth, green algae, waste tea leaves) into solid biofuel by hydrothermal carbonization at 260–300 °C; characterized the products by ultimate and proximate analysis, SEM, XRD, TGA/DTG, and higher-heating-value (HHV) measurement.
- Produced hydrochar and biocrude with HHVs of 27.3 MJ/kg and 22.98 MJ/kg at 300 °C, well above those of the raw feedstocks; published the results at ICMERE 2019 and extended them in *Waste and Biomass Valorization* (2025).

Collaborative Researcher: Carbon Capture, Clean Energy, and Energy Policy *2019–Present*

- Wrote a first-author review of CO_2 capture, transportation, utilization, and storage, now cited 63 times (*Environmental Science and Pollution Research*, 2025); also first author of CO_2-conversion and CO_2-separation studies in *Energy Reviews* (2026) and *Sustainable Chemistry for Climate Action* (2025).
- Proposed a compressor-driven system that captures CO_2 and SO_2 from industrial workplace air and reuses the CO_2 as a fire-extinguishing agent; the concept matured through three conference papers into *Separation Science and Technology* (2023).
- Simulated hydrothermal liquefaction of three microalgae in Aspen Plus: *C. vulgaris* biocrude reached 37.27 MJ/kg at 621 K, and *P. tricornutum* recovered 88.78% of feedstock energy (*Heliyon*, 2024).

Renewable Energy Resources and Its Optimization Research (collaborative simulation program) *2021–2024*

- Co-authored studies on renewable energy resources for Bangladesh and Nepal (30 and 26 citations), photovoltaic (PV) technoeconomics (4 citations), end-of-life PV recycling, e-waste management (25 citations), Electricity from rotational speed of train wheel (Conference proceeding) and EV-charging anomaly detection (*IEEE Access*, 2025).
- Co-authored studies on different optimizing PV system design by RETScreen, Homer and PVSyst softwares.

PROFESSIONAL & TEACHING EXPERIENCE

Assistant Engineer (Mechanical), Roads and Highways Department, Government of Bangladesh *Feb 2026–Present*

- Appointed to the mechanical engineering cadre through the 44th Bangladesh Civil Service (BCS) examination; support mechanical works for the agency that builds and maintains the national highway and bridge network.

Instructor, Directorate of Technical Education, Government of Bangladesh *Jan 2025–Feb 2026*

- Selected through the 43rd BCS (Non-Cadre, 9th Grade); taught mechanical engineering subjects and mentored diploma-level students in the national technical and vocational education system.

Industrial Intern, Haripur Power Plant Ltd. (360 MW combined cycle power plant), Narayanganj *May–Jun 2018*

- Rotated through the operation, maintenance, engineering, and safety departments; covered gas and steam turbine operation and heat recovery steam generation.

Intern, Microsoft Young Bangla Internship Program *Oct–Nov 2018*

- Selected for a national cohort on innovation, ideation, and technology entrepreneurship, run by Microsoft Bangladesh and the Centre for Research and Information.

PUBLICATIONS

230+ citations, h-index 9, and six Q1 journal articles (Google Scholar, July 2026). The applicant's name appears in boldface; the full list with DOIs is on Google Scholar.

Journal Articles (16)

1. I. Murshed, **M. S. A. Joarder**, T. Tanam, F. Islam, and M. Hasan, "Recycling of waste materials from solar photovoltaic panels: A Bangladesh perspective," *Environmental Progress & Sustainable Energy*, Early View, 2026.
2. **M. S. A. Joarder** and M. Mourshed, "Carbon capture and conversion: A potential game changer for clean energy," *Energy Reviews*, Art. 100192, 2026.
3. M. M. Islam, A. Rafat, F. Islam, **M. S. A. Joarder**, et al., "Optimizing PV system design with RETScreen: A comparative analysis of high-performance and lower collector area modules," *International Journal of Photoenergy*, Art. 3275702, 2026.
4. I. Murshed, F. Islam, **M. S. A. Joarder**, M. M. Islam, M. Y. Ali, and R. Ahshan, "Operational anomaly detection in EV charging systems using multi-parameter data analysis," *IEEE Access*, 14: 2552–2565, 2025.
5. **M. S. A. Joarder**, M. F. Kabir, M. H. Hasan, M. S. Islam, and A. Kutub, "CO2 separation technologies: Clean energy solutions for greening environment," *Sustainable Chemistry for Climate Action*, Art. 100168, 2025.
6. M. S. H. K. Tushar, F. F. Simanto, M. A. Hossain, M. S. Islam, and **M. S. A. Joarder**, "Simulation of hydrothermal carbonization for producing biofuels from water hyacinth and sewage sludge," *Waste and Biomass Valorization*, 16(12): 6637–6650, 2025.
7. A. Raihan, **M. S. A. Joarder**, S. M. Rahman, A. B. M. M. Bari, M. Ridwan, and T. Sarker, "Renewable energy resources for green development in Bangladesh: Perspectives, challenges, and opportunities," *Innovation and Green Development*, 4(5): 100298, 2025. (11 citations)
8. M. S. Islam, M. R. Hasan, K. Mostakim, **M. S. A. Joarder**, M. H. Hasan, and M. R. Ahmed, "E-waste management in Bangladesh: Environmental impacts, health risks, and sustainable policy strategies," *Cleaner Waste Systems*, 11: 100297, 2025. (25 citations)
9. **M. S. A. Joarder**, M. S. Islam, M. H. Hasan, A. Kutub, M. F. Kabir, F. Rashid, and T. A. Joarder, "A comprehensive review of carbon dioxide capture, transportation, utilization, and storage: A source of future energy," *Environmental Science and Pollution Research*, 32(15): 9299–9332, 2025. (63 citations)
10. F. Islam, M. Y. Ali, **M. S. A. Joarder**, et al., "Technoeconomic and environmental analysis of institutional grid-connected PV systems," *International Journal of Photoenergy*, Art. 5222213, 2025. (3 citations)
11. M. S. H. K. Tushar, M. S. Islam, T. Ahmmed, and **M. S. A. Joarder**, "Simulation of biocrude production from *P. tricornutum*, *S. platensis*, and *C. vulgaris* using Aspen Plus®," *Heliyon*, 10(17), 2024. (3 citations)
12. S. M. T. Rahman, **M. S. A. Joarder**, A. M. Hashan, S. M. M. Rahman, and R. H. Ali, "Holistic approach to safety and operational stability: Analyzing VVER-1200 reactor dynamics in SGTR and AC power loss," *Journal of Power and Energy Engineering*, 12(9): 102–130, 2024. (2 citations)
13. **M. S. A. Joarder**, "Analysis of the current development of renewable energy technologies in Bangladesh," *WSEAS Transactions on Environment and Development*, 2024. (sole author; 30 citations)
14. A. Raihan, **S. A. Joarder**, T. Sarker, B. Gosik, D. Kusz, and G. Zimon, "Renewable energy in Nepal: Current state and future outlook," *International Journal of Energy Economics and Policy*, 14(6): 395–407, 2024. (26 citations)
15. **M. S. A. Joarder**, F. Rashid, M. A. Abir, and M. G. Zakir, "A prospective approach to separate industrial carbon dioxide and flue gases," *Separation Science and Technology*, 58(10): 1795–1805, 2023. (9 citations)
16. S. I. Shojib, M. A. I. Sardar, **M. S. A. Joarder**, M. G. Zakir, and A. Hossain, "Heat transfer analysis of nanofluid based coolant used in the sub-channel of fuel assembly of a pressurized water reactor," *Journal of Nanofluids*, 12(2): 580–588, 2023. (4 citations)

Conference Papers (7)

17. S. Barua, M. Tabassum, **M. S. A. Joarder**, M. G. Zakir, M. A. Razzak, and M. A. Rahman, "A comparison of simulation analysis of safety systems and severe accident progression for VVER-1200 and AP-1000 following LOCA in hot leg by

PCTRAN," *Proc. Int. Conf. on Mechanical Engineering and Renewable Energy (ICMERE)*, Chattogram, Bangladesh, 2021. (4 citations)

18. **M. S. A. Joarder**, F. Rashid, M. G. Zakir, and T. A. Joarder, "Electricity generation from the rotational speed of train wheel: An approach of future energy," *Proc. ICMERE*, 2021. (7 citations)
19. S. Akter, **M. S. A. Joarder**, M. G. Zakir, A. Hossain, M. A. Razzak, and M. S. Islam, "Comparative analysis of thermal hydraulic parameters of AP-1000 and VVER-1200 nuclear reactor for turbine trip concurrent with Anticipated Transient Without SCRAM (ATWS)," *Proc. IEEE Int. Conf. on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, 2021. (11 citations)
20. S. F. Hassan, F. B. Farooq, M. M. Islam, M. G. Zakir, and **M. S. A. Joarder**, "Investigation on steam flow in a secondary circuit of a VVER-1000 power plant," *Proc. IEEE ACMI*, 2021. (2 citations)
21. M. S. H. K. Tushar, **M. S. A. Joarder**, R. F. Raj, M. A. Sarker, and M. R. Mahmud, "Hydrothermal carbonization and hydrothermal liquefaction of water hyacinth," *Proc. ICMERE*, Chattogram, Bangladesh, 2019. (10 citations)
22. **M. S. A. Joarder**, F. Rashid, and T. A. Joarder, "Solution to air pollution for removing CO2 and SO2 from flue gases: A prospective approach," *Proc. ICMERE*, 2019. (15 citations)
23. **M. S. A. Joarder**, T. A. Joarder, and F. Rashid, "Safety at industrial working place by separating carbon dioxide gas," *Proc. Int. Conf. on Mechanical, Industrial and Materials Engineering (ICMIME)*, Rajshahi, Bangladesh, 2019. (5 citations)

HONORS & AWARDS

- Champion, Poster Presentation, Inter University Science Festival 2017, RUET, for the poster "Ensuring Health & Safety at Industrial Working Place" (2017).
- GPA 5.00 with Golden in both HSC (2014; ranked 53rd, Jashore Board) and SSC (2012; Board Secondary Scholarship).
- Speaker at a mock TEDx event of the RUET English Club (2018); Certificate of Appreciation, Microsoft Young Bangla Internship Program (2018).
- Poster contest participant, Technovanza 2016 (North South University) and ESONANCE 2017 (Islamic University of Technology).

SCHOLARLY SERVICE & LEADERSHIP

- Peer reviewer for *Discover Applied Sciences* (Springer Nature), 2026.
- Presented six papers at four international conferences: **ICMERE 2019 and 2021, ICMIME 2019, and IEEE ACMI 2021.**
- Moderator and Head of Project Alokito Shishu Foundation, a volunteer organization that educates street children.
- Organizing Secretary (Admin), Society of Automotive Engineers, RUET; Joint Secretary, Greater Kushtia Association of RUET.

TECHNICAL SKILLS

Simulation & Modeling: Aspen Plus, PCTRAN, RETScreen, HOMER, MATLAB
Design & CAD: AutoCAD, SolidWorks (certified two-week training, SAE RUET, 2016), engineering drawing
Experimental Methods: hydrothermal carbonization and liquefaction; SEM, XRD, TGA/DTG, proximate and ultimate analysis
Programming & Tools: C (basic), data analysis, Microsoft Word, Excel, PowerPoint

TRAINING

- **Intelligent Transportation Systems (ITS):** Successfully completed training program on "Intelligent Transportation Systems (ITS)", gaining practical knowledge of modern technologies for smarter, safer, and more efficient transportation.
- **Orientation Training for Newly Appointed Assistant Engineers:** Successfully completed the orientation training for newly appointed assistant engineers at Roads and Highways Department (RHD), Bangladesh.

LANGUAGES

Bengali (native); English (full professional proficiency).

REFERENCES

Available upon request.